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GLA University, Mathura

Course: -B. Tech. II-Year, I-Mid Term (Odd Sem.) Examination, 2013-14

Subject: Data Structures and Algorithms (CSE-211)

Time: - 90 Minutes **M.M:-20**

Notes:-

1. Answer **ANY FOUR** questions from Section A, **ANY THREE** from Section B and **ANY TWO** from Section C.
2. All questions of the particular section should be answered collectively at one place.
3. Answer should be to-the-point and wherever required, be supplemented with neat sketches.
4. Any missing data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate marks.

Section-A (Attempt any **four** questions) 04 x 1 = 04

1. How many fields a Linked List node should contain for representing a polynomial and what are these?
2. Write the structure of a doubly linked list node using struct keyword.
3. Write two disadvantages of Linked List over an array.
4. Write two applications of Stack in the field of Computer Engineering.
5. Write the condition for checking a Queue for empty status.
6. Three algorithms do the same task. Time complexity of Algorithm 1 is $O(N^2)$, Algorithm 2 is $O(N)$ and Algorithm 3 is $O(\log_2 N)$ .Which of the algorithm should execute the fastest for large values of N and why?

Section-B (Attempt any **three** questions) 03 x 2 = 06

1. Explain Big O and Big Ω notations.
2. Convert the following arithmetic infix expression into postfix expression

   A+ (B*C-(D/E^F)*G)*H
3. Discuss Tower of Hanoi problem with a solution for three disks using neat diagrams.
4. Explain the algorithm for deletion for circular queue implemented using an array.

Section-C (Attempt any **two** questions) 02 x 5 = 10

1. What is Queue? Write an algorithm to insert an element in Queue using Linked List.
2. Write an algorithm to evaluate a postfix expression using Stack.
3. Write a C Program to implement addition of two polynomials using linked list.

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GLA University, Mathura

Course: -B.Tech II-Year, I-Mid Term(ODD Sem.) Examination, 2013-14

Subject: - Computer Organization (CSE-212)

Time:-90 Minutes **M.M:-20**

Notes:-

1. Answer **Any Four** questions from Group A , **Any Four** from Group B and **Any Four** from Group C.
2. All questions of the particular group should be answered collectively at one place. All Parts of a question (a, b, etc.) should be answered at one place.
3. Answer should be brief and to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate full marks.

Group-A (Answer any Four) 1x4=04

I. Subtract $(7250)_{10}$ - $(250)_{10}$ using (r-1)'s Complement.
II. What would be the bias value for a base-2 exponent in a 6-bit field?
III. How many parity bits are required with 5 data bits in hamming code?
IV. Explain the representation of NaN.
V. Perform addition in binary using 2's complement representation (+52) and (-13).
VI. Consider two normalized floating point numbers 0.9504×10^3 and 0.8200×10^2. Find their product.

Group-B (Answer any Four) 1.5x4=6

I. Express (-105) in IEEE 32-bit floating point format.
II. Draw the flowchart of main functions required for addition and subtraction for floating point numbers.
III. Discuss advantages and disadvantages of General-register based processor organization.
IV. Explain the procedure to implement binary multiplier.
V. Discuss the various types of register used in computer system.
VI. Perform binary multiplication of 101 x 110.

Group-C (Answer any Four) 2.5x4=10

I. Find the expression for generating carry in 4-bit carry look ahead adder. Draw the logic diagram also.
II. Suppose an 8-bit data word stored in memory is 11000010. Using Hamming algorithm, determine what parity bits would be stored in memory with the data word. Suppose while transmitting the message a change in bit position 9 has occurred. How the error is detected and corrected at receiver end.
III. Show the step by step multiplication process using Booth's algorithm when the following numbers are multiplied. Assume 5-bit register that hold numbers (+15) * (-13).
IV. Write a program to evaluate an arithmetic expression A= (B+C)*(D+E)/F using:
    i. Three- address instruction
    ii. Two-address instruction
    iii. One-address instruction
    iv. Zero-address instruction
V. What is an instruction cycle? How can interrupt affect it? Explain clearly.
VI. Consider a two word instruction "LOAD TO AC" stored at location 200 and 201 with address field equal to 500. Content of PC has value 200(for fetching the instruction). Content of processor register R1 is 400. Content of an index register XR is 100. After the instruction is executed, the accumulator AC receives the operand. For each possible mode, find the effective address and the operand that is to be loaded into AC.

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GLA UNIVERSITY, MATHURA

COURSE: B.TECH. II- Year, I- Mid Term (Odd Sem.), Examination, 2013-14

Sub: Object Oriented Programming (ITE 211)

Time: - *90* Minutes **M.M.:-*20***

1. Answer **ANY FOUR** questions from each section.
2. All questions of the particular section should be answered collectively at one place.
3. Answer should be to-the-point and wherever required be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate full marks.

Section A (Attempt any four questions) ***4*1=4***

Q1. What are command line arguments?
Q2. Name the principles of OOP.
Q3. Explain Access modifiers in JAVA.
Q4. What are Instance variables in JAVA?
Q5. Define Information hiding.
Q6. What is Class and Object?

Section B (Attempt any four questions) ***1.5*0.4=6***

Q1. What is modeling? What are the different views in UML?
Q2. What do you mean Use Case? Design a Use Case diagram for student registration system.
Q3. What do you mean by constructor? Explain with a suitable example.
Q4. Design Class diagram for following:
Companies may employ many people, and people may work for many companies. Every employee in a company has a manager who may manage many subordinate employees. Show the relationship between the Employee and Company classes in UML.
Q5. What is sequence diagram? When is it appropriate to use this diagram?
Q6. Write a program to generate the Fibonacci series using parameterized Constructor.

Section C (Attempt any four questions) ***04*2.5=10***

Q1. What is Object Orientation? Explain advantages of Object Oriented approach of system design over structured approach of system design.
Q2. What are class diagrams? Explain how classes are identified in designing of an object oriented system, with the help of an example.
Q3. Explain the following terms:
A. Generalization B. Association C. Multiplicity D. Qualified Association E. Aggregation
Q4. What is UML? Explain the Conceptual Model of UML.
Q5. What are the features Of JAVA? Explain in details.
Q6. What is Polymorphism? Explain with the suitable example.

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GLA University, Mathura

Course – B. Tech., II-year, I-Mid Term (Odd Sem) Examination, 2013-14

Subject – Discrete Mathematics (CSE-213)

MM: 20

Time – 90 minutes

Note:

1. Answer any four questions from Section A and Section B, and any two questions from Section C.
2. All questions of a particular section should be answered collectively at one place.
3. Answer should be to-the-point and wherever required, be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate marks.

Section A (Attempt any four)

(4x1=4)

I. The probability of a horse winning a race is 1/5 and the probability of a horse B winning the same is 1/4. What is the probability that none of them wins?

II. Find the generating function for the sequence 1, 0, 1, 0, 1, 0, 1, 0 .....

III. How many distinct relations can be made on a set of 3 elements?

IV. If x and y are real numbers then max (x, y) + min (x, y) is equal to .....

V. Out of 80 students in a class, 60 play football, 53 play hockey, and 35 play both the games. How many students do not play any of these games?

VI. Given that the ordered pairs (a, 7) and (-4, b) belong to the relation $\{(x, y) : 2y-3x = 8\}$. Find the values of a and b.

Section B (Attempt any four)

(1.5x4=6)

I. Prove that : $A - (B - C) = (A - B) \cup (A \cap C)$

II. Find a simple expression for the generating function for the following numeric function:
1, -2, 3, -4, 5, -6, 7, -8, ...

III. What is the probability that Abby, Barry, and Sylvia win the first, second and third prizes, respectively, in a drawing if 200 people enter a contest and
   a) No one can win more than one prize.
   b) Winning more than one prize is allowed

IV. Among the positive integers less than or equal to 100, let A be the set of even integers and let B be [redacted] set of integers divisible by 3. How many integers are neither even nor divisible by 3?

V. Determine the values of the constants A and B so that $a_n = An + B$ is a solution of the recurrence relation $a_n = 2a_{n-1} + n + 5$.

VI. Define countable and uncountable sets with examples.

Section C (Attempt any two)

(2x5=10)

I. Let A denote a set of real numbers. A relation R is defined on A such that (a, b) R (c, d) iff $a^2 + b^2 = c^2 + d^2$ Show that R is an equivalence relation. Also determine whether the relation $S = \{(a, b) \in S \mid (a-b)<=3\}$ is reflexive, symmetric and/or transitive.

II. Solve the following recurrence relation using the method of characteristic equation.
$a_n - 6a_{n-1} + 8a_{n-2} = 4^n$ given $a_0=8$, $a_1=22$

III. Solve the recurrence relation by the method of generating functions $a_n - 9a_{n-1} + 20a_{n-2} = 2$ for $n \geq 2$ with initial conditions $a_0 = 1$, $a_1 = 2$.

GLA University, Mathura
I Mid-Term Examination, 2013-14

Course: - B.Tech. **II-Year, III-Sem.**

Subject: - Mathematics (AHM- 201) **Uni. Roll No:-**

Time:-1Hour 30 Minutes **Total Marks:-20**

Notes:-

1) Attempt **ANY FOUR** questions of Section- A, **ANY THREE** from Section- B, and **ANY TWO** from Section- C.
2) Answer should be brief and to-the-point and be supplemented with neat sketches.
3) Any missing or wrong data may be assumed suitably giving proper justification..

Section - A (1 × 4 = 04)

Q .1 State Gauss divergence theorem.

Q.2 If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then find directional derivative of $\frac{1}{r}$ along $\vec{r}$.

Q.3 Find the unit tangent vector to the curve $\vec{r} = t^2\hat{i} + 2t\hat{j} + t^3\hat{k}$ at $t = 1$.

Q.4 Find all the roots of the equation $x^3 + 6x^2 + 9x + 4 = 0$, if one root of equation is -1.

Q.5 Find the equation whose roots are decrease by 2 to roots of the equation $x^3 - 6x^2 + 12x - 8 = 0$.

Section - B (2 × 3 = 06)

Q1. Solve the equations $x^3 - 27x + 54 = 0$ by Cardon's method.

Q.2. Solve the equations $x^4 + 8x^3 + 9x^2 - 8x - 10 = 0$ by Descarte's method.

Q.3. Using Green's theorem, evaluate $\int_C (x^2 ydx + x^2 dy)$ where C is the boundary described anti-clockwise of the triangle with vertices (0, 0), (1, 0), (1, 1).

Q.4. Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $x^2 + y^2 - z = 3$ at the point (2, -1, 2).

Section - C (5 × 2 = 10)

Q.1 Verify Stoke's theorem for $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\,\hat{j}$ taken round the rectangle bounded by the lines

$x = \pm a, y = 0, y = b$.

Q.2 Solve the equations $x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$ by Ferrari's method.

Q.3 If the directional derivative of $\phi = ax^2 y + by^2 z + cz^2 x$ at the point (1, 1, 1) has maximum magnitude 15 in the direction parallel to the line $\frac{x-1}{2} = \frac{y-3}{-2} = \frac{z}{1}$, find the values of a, b, c.

Q.4. Show that the vector $\vec{F} = \frac{\vec{r}}{r^3}$ is irrotational as well as solenoidal, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = \sqrt{x^2 + y^2 + z^2}$. Find the scalar potential.

[redacted]

GLA University
Ethics & Values (AHE: 201)
First Term Exam (2013-14)
B. Tech: II Year, 3rd semester |CSE, EC, E.E & E.N|

Marks: 20 **Time: 1.30 Hours**

This paper consists of two passages. Carefully read the passages and answer the following multiple choice questions.
Full marks (assigned) will be given for a complete answer that means all correct choices.
Half mark (assigned) will be given for an incomplete answer that means one or more correct choices but not the all.
Zero (0) will be given if one clubs an incorrect choice with one or more correct choices.
Marks are assigned before passages.

Passage-1 (1 X10 = 10 Marks)

Morality offers an explanation to the questions: what is right and wrong, good and bad, and what ought to be done or what ought not to be done? In other words, morality provides the guidelines for our conduct, choices, actions, and decisions. What should be the way of life, and in what manner we should live in the society? These are some significant questions which we examine in the moral discourses. However, some scholars maintain the fact that morality cannot be taught. It is just a matter of practice. But, we can have a meaningful discourse or a conversation on the moral issues and questions. Although moral guideline gives direction, it should reflect in our routine practices. Let's try to understand the structure of morality. Analyzing the structure of morality on superficial level or at the most outer core of morality, we find the relationships. It means that morality begins with the understanding of our relationships in our social settings. When we locate ourselves in the society we find ourselves related with some kind of moral binding. For instance being a student you are a daughter or son of your parents, brother or sister of your sibling, a student of a teacher and a friend of your fellow-beings. However, your being in relationship with the others gets its meaning by understanding of your roles. By roles we understand our participation in our relationships; the way we act in a certain manner. But, unless we effectively acknowledge our rights and act on our duties we cannot competently play our roles in a justified sense of morality. Here, arises a question: what is justified sense of morality? Well, justified sense of morality means to be abided with the moral rules and follow them when and wherever we require in a certain social circumstances. Hence, morality is well grounded in the moral rules. The rules are meant to follow in our mundane life. By the rule following not only we well organize our conduct but also develop a sort of discipline and maturity that we require to live our life in a better way. However, one obvious question at this point is: where do the moral rules spring from? The values are the originating source of moral rules. For instance take any moral let's say: Keep your promise. Promise keeping is a moral and when you do it you are doing the right thing and your behavior is appropriate. But, the promise keeping, the moral, is based on the trust that is a value. Therefore, values are fundamental to our moral practices. Moreover, in the value assessment we examine the morality and we investigate into the conduct, character and decisions of a person.

Generally, people don't differentiate between the ethics and morality. They use these two terms interchangeably. But, there is a subtle difference between the two. And, the difference can be grasped well when we distinguish between the ordinary morality and the qualified morality. Let us say somebody is following social rules and norms without inquiring into them. S/he may be living the life without facing any grate trouble. And, s/he is having faith in morality. However, if one follows a rule or a norm without knowing the meaning of the rule or norm then it can be misleading; as there is no guarantee that if a rule or norm is followed by the majority then it will be absolutely right and proper. Moreover, suppose someone is ordinarily following the moral conduct by suppressing his/her own freedom, interests or desires. Can his/her conduct be said to be morally proper in true sense? No. And, the response makes it clear that morality in a qualified

sense does not mean just following. What it means is having critical acceptance of morality. In this way, morality in a truly qualified sense differs from that of just following ordinary morality. The qualified sense involves one's critical attitude towards morality. When we raise certain questions on the morals, values and socio-cultural systems which exist in the first-order, then we move from first-order to the second order of ethics. And, in this way we understand ethics as a second order critical inquiry in which we examine the first order of morality by raising certain questions, putting forward certain ideological and theoretical viewpoints. However, ethics most importantly involves one's critical point of view or what we call one's moral point of view. The understanding of ethics amounts mature thinking of people on the questions of rights and wrongs. The so called maturity demands one's rational and autonomous critical attitude. In simple terms, I am being ethical or moral in a qualified sense only when my conduct is grounded in reasons, my conduct is freely governed by myself, and my conduct represents my true intentions. This kind of moral conduct is really meaningful in ethical sense because it really belongs to someone, and can be regarded as a genuine case of moral responsibility. Because, it is only in this sense we can hold someone responsible for his/her proper or improper behavior. And, think, if someone's conduct is grounded in a social threat, pressure, and compulsion. Can it be called moral? Response is undoubtedly flat: No. Having set this clarity about the morality (qualified sense) in true sense now we can understand it as a system of right codes of conduct we critically design and practice. Thus, morality by defining what is right and good in a particular situation help us to make the choices and perform the actions in this world. Morality defines what ought to be done to live the social life in a proper fashion.

Justice means fare distribution of the resources, rights, honors, opportunities and educational seats in a socio-political-economic system. However, for some it is not only a distributive notion but also has a purpose to create a well ordered society. There are three prominent approaches to justice. First, this approach is grounded in individual freedom and argues that justice is to provide full freedom to an individual to live his/her life. In any case individual rights should not be sacrificed for any wider social goal. This is why the view is individualistic as it gives importance to the individuals over society. The second approach is of the view that society is more important than individuals, and hence justice is to maintain maximum good for the maximum numbers. This view is grounded in social welfare and makes the point that we should sacrifice the individual rights if they disturb the interests of maximum people. This view does not care the minority rights. It gives priority to the society. The third approach is based on the theory of virtue that the purpose of justice is to create a well ordered society by rewarding people according to their virtues: qualities, capacities and merits. This view does not maintain the priority. It points out the fact that by respecting virtues not only we can avoid many conflicts between individual and society but also can respect freedom of individuals as whatever they choose will be according to their strengths and skills. The virtue view is also a socialist point of view. However, it differs from the social welfare approach in that it does not put emphasis on the just distribution and priority of society.

1) What do you mean by the ordinary Moral conduct?
   - One's just following attitude.
   - One's unquestioning moral behavior.
   - One's critical moral attitude.
   - One's rule following.
2) Governed by the ordinary kind of morality one's moral conduct cannot be proper in true sense of morality. What does the statement mean?
   - One may be acting under pressure.
   - One may be acting under a social threat.
   - One may be acting without one's interest.
   - One is acting with freedom.
3) Which of the following statements are true about the moral values?
   - Moral rules are grounded in values.
   - In absence of them we cannot have any meaningful talk of morality.
   - We assess one's character through values.
   - Moral rules are fundamental to our moral practices.
4) How does the qualified true sense of morality differ from that of ordinary sense?
   - True sense of morality is grounded in one's rational attitude.
   - True sense of morality is grounded in faith.
   - True sense of morality represents one's true intentions.
   - True sense of morality is one's moral conduct governed by one's freedom.

5) In what sense do we understand ethics a second order inquiry?
   - ➢ It is a critical investigation into the first order of morality.
   - ➢ It tests morality by raising questions.
   - ➢ It examines a moral by respecting one's own point of view.
   - ➢ None of these.
6) Which of the following statements are not true about the justice?
   - ➢ It is just distribution of resources, rights and opportunities.
   - ➢ It is only a distributive notion.
   - ➢ It has a purpose to create a well ordered society.
   - ➢ Justice has only purpose to respect freedom.
7) If an engineering student is following the wish of his/her parents to become an engineer not on the basis of faith in them but by applying reason, interests and freedom of his own. How would you grade his/her conduct?
   - ➢ A just following conduct.
   - ➢ Moral conduct in true sense.
   - ➢ Right and reasonable conduct.
   - ➢ Free but immoral conduct.
8) What is justified sense of morality?
   - ➢ We competently participate in our roles.
   - ➢ We acknowledge our rights and act on our duties.
   - ➢ We are morally abided by the moral rules.
   - ➢ We follow the moral rules.
9) What are the problems of social-welfare approach?
   - ➢ It does not care the rights of minority.
   - ➢ Individual freedom is not guaranteed.
   - ➢ In cares only the good of maximum number.
   - ➢ It is a socialist view.
10) Which of the following points are not right about the virtue approach?
   - ➢ It gives priority to the society.
   - ➢ It respects one's freedom by respecting one's strengths and skills.
   - ➢ It puts emphasis on the purpose of justice.
   - ➢ It puts emphasis on the just distribution.

Passage: 2 (1X10 = 10 Marks)

What could be the most challenging other than making a decision? Indeed, decision making is always a tough task but deciding in a situation where the choices are neither completely right nor completely wrong; the decision making gets tougher. The qualified sense of morality or the ethical view point is very much necessary for a decision maker. It may be the fact that someone decides through certain reasons however does not think from an ethical point of view whether his/her decision would affect someone or not. And, this is the reason why an ordinary decision making differs from that of ethical decision making (E.D.M.). In ethical decision making we consider not only the desired results but also the rightness and wrongness of our actions on the part of our decisions. In decision making typically one is tasted in some pressing situations. However, the key factors of decision making are rationality and critical thinking. Let us talk about one hypothetical case namely, the Trolley Car. Hypothetical cases involve some constructed situations in order to test one's real decision making and problem solving skills. In this case as we all know there are three different versions that actually point out the variations in our decision making with the changes in the conditions. However, the case of trolley car shows that our decision making is not simply based on reasons but also amounts to certain moral emotions which in some sense can be said to be reasonable in so far as they carry some right reasons. In others words, we can say sometimes our emotions are proper, right and justified. In this way sometimes our emotional response can be right and reasonable. In the first version of the trolley car case: Suppose you are the driver of a trolley car hurtling down the track at sixty miles an hour. Up ahead you see six workers standing on the track, tools in hand. You try to stop, but you cannot. The brakes do not work. You feel desperate, because you know that if you crash into these six workers, they will all die. Suddenly you notice a side track off to the right. There is a worker on the track, too, but only one. You realize that you can turn the trolley car onto the side track, killing the one worker, but saving the six. How would you respond in this situation as a driver? Majority would respond that they turn the trolley with the reason that it would spare more lives. Taken for granted that the worker on the side track is innocent majority would say six men on the main track are also innocent. And, the decision would be based on the reason that we should go for the better consequences. However, the reason is subtle in that it involves something more than just better consequences and it is about the nature of choice itself as the choice is not objectionable and unreasonable because situation is confined and we cannot do anything otherwise. However, one issue remains a matter of debate: How can a person take decision on the other's life? Moving to second

version when trolley car is automatic and there is no side track. You are standing on the bridge as an onlooker. Down the track comes a trolley, and at the end of the track are six workers. Once again breaks do not work and trolley car is about to crash into those six men. You feel helpless to prevent this disaster. You think jump on the track but finding yourself too thin to stop the trolley you change the mind. However, you notice alongside you a very fatty man just leaning onto the bridge. You reasonably think that if you give him a gentle push he will fall on the track before the trolley car and will obstruct the track anyhow. Although the fatty man will die, he would save the life of six men. Majority would decline the choice to give a gentle push to fatty man for saving six down on the track. The choice is not like the first version because this version is a bit open and free in that sense the onlooker has got option not to involve himself. In this version the onlooker is not directly in the situation like the driver in the first case. But, why the majority is not minding the better consequences in this version because if we compare the consequences they are same six lives saved at the cost of one. However, we people all the time do not mind consequences only. If the nature of an action is not objectionable we simply mind the consequences but if the nature of an action involves wrongness or cruelty of a kind we mind first the nature of an action itself and more often decline to go for better consequences. However, if throwing fatty man involves direct cruelty then is it not true that letting six people die shows our passive cruelty that we witness them dying in front? Thinking reasonably one can understand that the so called passive cruelty allows more cruelty to take place. Well, the other reason could be offered in favor of saving the fatty man that in the first case driver could not had his intention to kill the innocent standing on the side track because driver had no choice other than moving the trolley on the given two tracks; either go straight ahead or turn. But, in this second version onlooker has this choice whether to push the fatty man or not. And, it is in this way he can be intentional in both ways either he pushes or not. However, fatty man is not the part of this situation because he is unknown to both the onlooker and his decision. However, the onlooker is not directly involved in physical terms, but is very much directly involved in so far he can have the intention to quiet pushing the fatty man and letting the six people die. Sometimes our quitting attitude shows the lack of moral courage or our emotional arrest that does not allow wider reason. Sometimes, we simply mind our own interests and do not care about humanity in a wider sense. However, the third version of this case in which there is a change in terms of condition that we need not to push the fatty man instead we are given a button that is operating a trap onto which the fatty man is standing on. We have to press the button. It will open the trap and fatty man will fall down on the track before the trolley car. He will obstruct the track and it would save the life of six men. Majority cannot be so reluctant and passive in this case as it was in pushing the fatty man. As a result, majority will be divided in pressing or not pressing the button. What is the reason that actually divided the majority and allowed more number of persons to save the six men by pressing the button? The reason for the person, who goes with moral courage and applies a wider reason beyond his own interest of saving more people, is that the third version involves less guilty factor than the second, because if a person has this compulsive intention to kill one man in all the three versions then the person has this more reasonable and wider intention to save the six. The factor of guilt is decisive for the person who really wants to save six. And, the person who really saves the six in all three versions have least guilt in pressing the button, a bit more guilt in turning the trolley and s/he would feeling most of guilty in the pushing the fatty man down on the track. Thus, indeed the means and method matter for the person who only kills one worker unintentionally or having compulsive intention. As guilty is a true sense of our morality it is directly attached with our intentions and realization of wrongness. We would prefer to commit least wrong actions in order to minimize the guilty sense.

1) Why majority would decline to push the fatty man?
   - It involves direct involvement.
   - Majority has choice not to involve.
   - It involves indirect involvement.
   - None of these.
2) Which of the following statements are true about Ethical Decision Making (E.D.M.)?
   - It carries ethical point of view.
   - It carries reasonable consequences.
   - It checks whether the action is right or wrong.
   - It carries only desired consequences.
3) Which of the following statements are not true about the second version of trolley car case?
   - Majority would turn the trolley car.
   - Majority is not confined like turning the trolley car in the first case.
   - Majority would push the fatty man.
   - Majority would decline to save six men.

4) Guilt is true sense of morality. What does author mean to this statement?
   - Guilty sense refrains from immorality.
   - Guilt is directly attached with our intention.
   - Guilt is one's realization of wrongness.
   - Guilt is attached with others.
5) What are the key factors of decision making?
   - Emotional response
   - Rationality
   - Critical-thinking
   - Self-interest
6) How do the three versions of trolley car case differ from one another?
   - On the grounds of reasons.
   - On the grounds of different conditions.
   - On the grounds of intentions.
   - On the grounds of better consequences.
7) Why do more number of people would incline to press the button?
   - They can simply press the button.
   - Pressing the button involves less guilty.
   - They would like to save six men.
   - None of these.
8) What would be the intention and the choice of the person who goes to save the six men in all versions?
   - Have compulsive intention to kill one.
   - Have intention to kill one.
   - Have compulsive choice to kill one.
   - Have intention to save six.
9) Which of the following statements are true about the trolley car case?
   - It tests problem solving skills.
   - It tests decision making skills.
   - It is about ethical decision making.
   - It is a real case.
10) We should not push the fatty man as pushing fatty man involves direct cruelty. But, why does not this reason seem undoubtedly true?
   - Fatty man was in the situation as he knows the onlooker.
   - It may be the case that someone lacks moral courage.
   - Quitting from direct cruelty and not pushing the fatty man will allow more cruelty in fact.
   - It may be the case that someone is under the arrest of emotions.